

A
MINI-PROJECT REPORT On
WOMEN SAFETY GLOVES

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

Submitted by

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WOMEN

(An Autonomous Institution)

(Affiliated to Jawaharlal Nehru Technological University Hyderabad, Accredited by NBA, NAAC with A+)
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(2025-2026)



VIGNAN'S INSTITUTE OF MANAGEMENT AND TECHNOLOGY FOR WOMEN

(An Autonomous Institution)

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CERTIFICATE

This is to certify that the project work entitled “**Women safety gloves**” submitted by **K.Sravanthi(23UP1A0529),Y.Mayuri(23UP1A0561),A.Tejasri(24UP5A0502), Samreen(23UP1A0553)**, in the partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, **VIGNAN'S INSTITUTE OF MANAGEMENT AND TECHNOLOGY FOR WOMEN** is A record of Bonafide work carried by them under my guidance and supervision. The results embodied in this project report have not been submitted to any other University or institute for the award of any degree.

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DECLARATION

We hereby declare that the results embodied in the project entitled “**Women Safety Gloves**” is carried out by us during the year 2024-2025 in partial fulfilment of the award of **Bachelor of Technology in Computer Science and Engineering** from **Vignan's Institute of Management and Technology for Women** is an authentic record of our work under the guidance of Guide Name. We have not submitted the same to any other institute or university for the award of any other Degree.

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Our sincere thanks to all the teaching and non-teaching staff of Department of Computer Science and Engineering for their support throughout our project work.

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ABSTRACT

Women safety is a major concern in today's society, especially during travel or isolated conditions. This project proposes a Smart Women Safety Gloves system that provides immediate assistance during emergencies. The gloves contain an SOS activation mechanism. It sends an emergency alert along with the user's live location to preregistered contacts through GSM/Bluetooth communication. Ensuring the safety of women in public and private spaces has become a critical social challenge due to the increasing number of harassment and assault cases. The Women Safety Gloves offer an effective, affordable, and practical solution for women's safety in modern society. Additionally, the gloves incorporate shock resistant materials to safeguard against physical harm. By combining comfort, safety, and smart technology, this project aims to empower women with a reliable, discreet, and effective tool for personal security. The innovation not only enhances confidence but also contributes to broader efforts in promoting women's safety and independence. Ultimately, the development of women's safety gloves reflects a broader commitment to equity in workplace safety. Their adoption not only safeguards health but also empowers women to thrive in diverse professional environments.

1.INTRODUCTION

1.1 Motivation

In recent years, the safety of women has become a critical issue due to the rising number of harassment and assault cases in both urban and rural areas. Many women face insecurity while traveling alone, especially during late hours or in isolated places. Despite advancements in technology, there is still a need for a simple, reliable, and immediate solution that can help women protect themselves and seek help quickly during emergencies.

The motivation behind developing women safety gloves comes from the need to provide a practical and wearable safety device that ensures instant response in dangerous situations. Unlike mobile-based applications, which may not always be accessible during panic situations, a glove-based system is easy to use and can be activated quickly with minimal effort.

1.2 Problem Definition

Women's safety remains a major concern due to the increasing number of harassment, assault, and crime incidents in public and private spaces. Many women feel unsafe while traveling alone, especially at night or in less crowded areas. In emergency situations, it is often difficult for them to quickly access help or inform others about their location. The problem surrounding women's safety gloves arises from the fact that most existing protective gloves in the market are designed with a generic or malecentric approach, which fails to adequately meet the ergonomic and safety needs of women. Standard gloves often do not fit smaller hand sizes properly, leading to discomfort, reduced dexterity, and compromised grip, which in turn affects precision and efficiency in tasks. This poor fit can also result in safety hazards, as loose or oversized gloves are more likely to slip, tear, or fail to provide sufficient protection against workplace risks such as cuts, abrasions, chemical exposure, or heat. Furthermore, the limited availability of gloves specifically tailored for women highlights a gap in inclusivity within personal protective equipment (PPE), leaving women with fewer options in industries where safety gear is mandatory.

Challenges

The women's safety gloves project comes with several challenges that span across technical, practical, and social dimensions. One of the foremost issues is power management, since wearable devices often rely on small batteries that may not last long enough during emergencies. Ensuring efficient energy use and exploring alternatives like solar or kinetic charging is crucial. Another challenge lies in hardware integration and miniaturization—fitting sensors, GPS modules,

communication chips, and encryption systems into a compact glove without compromising comfort or durability is complex.

1.Data security and privacy: It is also pose significant hurdles, as the gloves transmit sensitive information such as location and distress signals. Strong encryption and secure key management are essential, but implementing them on low-power microcontrollers can be difficult. Connectivity reliability is another concern, because alerts must reach recipients even in areas with poor network coverage. Without stable communication, the system risks failing when it is needed most.

2.Comfort and usability: These are critical. The gloves must be lightweight, durable, and easy to operate, while avoiding accidental triggers that could cause false alarms. Cost and accessibility present further challenges—advanced features may increase production costs, making the gloves less affordable for widespread adoption. Additionally, user training and awareness are necessary to ensure that women know how to use the gloves effectively.

3.Integration with emergency services: It requires collaboration with authorities, which can be logistically and bureaucratically challenging. Ensuring that alerts are not only sent but also acted upon promptly is vital to the project's success. Together, these challenges highlight the need for careful design, testing, and refinement to transform the gloves from a prototype into a reliable, scalable solution that truly enhances women's safety.

1.3 Objective

The main objective of the women safety gloves project is to provide safety and protection for women during emergency situations. The device is designed to send alert messages quickly to family members or friends and share the user's location using GPS. It aims to be simple, easy to use, and wearable so that it can be activated instantly when needed. The system also includes features like a buzzer or alarm to attract attention and get help from nearby people. We want achieve the following objectives:

- 1.To ensure women's safety.
- 2.Develop a reliable device that helps women in emergency situations.
- 3.To provide quick emergency response.
- 4.Enable instant activation of alerts with a simple action like pressing a button or applying pressure.
- 5.To send real-time location tracking.

1.4 Scope of Project

The scope of the women safety gloves project focuses on developing a smart and wearable safety device that can be used by women in emergency situations. The system is designed to provide instant alert messages and share real-time location details with predefined contacts using GPS and communication modules.

1. Objectives

- Design and develop safety gloves tailored specifically for women.
- Ensure protection against workplace hazards (chemical, mechanical, thermal).
- Provide comfort, proper fit, and ergonomic design for prolonged use.
- Promote safety awareness and accessibility for women in industrial and domestic settings.

2. Deliverables

- Prototype designs of women's safety gloves in multiple sizes.
- Material testing reports (durability, cut resistance, heat resistance, etc.).
- Final product samples ready for pilot distribution.
- Packaging and user guidelines emphasizing safety and usability.

3. Tasks & Activities

- Market research on women's safety gear needs.
- Material selection (breathable fabrics, reinforced layers, eco-friendly options).
- Ergonomic design studies for hand sizes and shapes.
- Safety compliance testing (ANSI, EN standards).
- Feedback collection from women users in different industries.

4. Timeline

- Research & design: 2 months
- Prototype development: 3 months
- Testing & certification: 2 months
- Final production & launch: 3 months

5. Resources

- Design team (product designers, ergonomics experts).
- Safety engineers for compliance testing.
- Manufacturing partners.
- Budget allocation for R&D, testing, and marketing.

6. Boundaries (Out of Scope)

- Development of other PPE (helmets, masks, etc.).
- Marketing campaigns beyond product launch.
- Customization for men's gloves (focus remains on women's safety).

7. Constraints

- Must comply with international safety standards.
- Budget limitations for material sourcing.
- Time-bound launch to align with safety awareness initiatives.

8. Acceptance Criteria

- Gloves meet safety certifications.
- Positive user feedback on comfort and usability.
- Successful pilot launch with measurable adoption.

This scope ensures the project is **focused on women's safety needs**, avoids scope creep, and sets clear deliverables.

2 LITERATURE SURVEY

Women's safety has become an important area of research due to the increasing number of crimes and unsafe situations faced by women. Many researchers have proposed different technological solutions to improve safety, especially using Internet of Things (IoT) and wearable devices. Studies show that smart safety devices use sensors such as pressure sensors, pulse sensors, and motion sensors to detect emergency situations and trigger alerts.

IJERS +1:

Several existing systems are designed with components like GPS and GSM modules to send real-time location and alert messages to family members or authorities. These devices are often built using microcontrollers such as Arduino or Raspberry Pi, which help in processing signals and activating emergency responses.

Research Gate:

Some research works have also introduced advanced features like automatic danger detection, audio recording, and image capturing to provide more security. In certain systems, alerts are triggered automatically when abnormal conditions are detected, reducing the need for manual activation.

IJERT:

Wearable safety devices such as smart bands, watches, and gloves are widely studied because they are easy to use and can be carried at all times. These devices aim to provide quick response, real-time tracking, and communication during emergencies. Several Morse Code translation systems currently exist, ranging from simple online tools to more advanced software applications. These systems vary in terms of functionality, user interface, and the platforms they support.

3.SYSTEM ANALYSIS

3.1 Existing System

Existing women's safety glove systems are typically IoT-based wearables that combine sensors, microcontrollers, and communication modules to detect emergencies and send alerts. Some also integrate self-defense mechanisms like electric shocks. These systems aim to provide discreet, portable protection for women.

Mobile App Integration:

- Examples: bSafe, Himmat, Raksha, MySafetipin, 112 India.
- Functionality: SOS alerts, live tracking, emergency contact notifications.
- Strength: Widely accessible, government-backed in some cases.
- Limitation: Requires manual phone use, network, and battery

Proto type Devices:

- Shock or alarm systems: Gloves that can emit sound or mild electric shocks to deter attackers.
- Biometric monitoring: Heart rate or stress detection to trigger alerts.
- Wearable IoT frameworks: Designed to empower women in public spaces by providing discreet, wearable safety solutions.

3.1.1 Disadvantages of Existing System

While existing Morse Code translators—whether online tools, mobile apps, desktop software, or browser extensions—provide basic functionality, they often suffer from several limitations.

- **Poor fit:** Most gloves are designed for men, so women often get loose or uncomfortable sizes.
- **Reduced dexterity:** Ill-fitting gloves make it harder to grip or handle tools safely.
- **Skin issues:** Long use can cause sweating, irritation, or allergies (especially with latex).
- **Limited options:** Few gloves are made specifically for women's hand shapes.
- **Narrow focus:** They mainly protect against workplace hazards, not personal safety concerns like harassment.
- **Durability issues** – Some gloves wear out quickly, especially cheaper ones, reducing longterm protection.
- **Limited style choices** – Few designs are made with women in mind, often ignoring aesthetics or preferences.
- **Reduced flexibility** – Thick protective materials can make hand movements stiff and awkward.
- **Cost factor** – Specialized gloves (cut-resistant, chemical-proof) can be expensive, making them less accessible.

- **Lack of awareness** – Many women are not informed about proper glove options, leading to unsafe substitutes.

3.2 Proposed System

A proposed system for women's safety gloves should be designed with ergonomics, inclusivity, and advanced protection at its core. Unlike existing gloves that are often scaled-down versions of men's designs, the new system would be based on anthropometric data specific to women's hand shapes, ensuring proper fit, comfort, and dexterity. The gloves should incorporate lightweight yet durable materials that provide resistance against cuts, chemicals, and abrasions while maintaining flexibility for fine motor tasks. To address health concerns, the system could use hypoallergenic, breathable fabrics with moisture-wicking properties to reduce sweat buildup and skin irritation. Smart features such as **sensor integration** for detecting hazardous substances or alerting users in unsafe conditions could further enhance safety. Additionally, offering a range of sizes and styles would empower women across different industries to choose gloves that suit their work environment without compromising protection. Overall, the proposed system emphasizes fit, comfort, safety, and inclusivity, creating PPE that truly supports women in the workplace:

OVERVIEW OF PROPOSED SYSTEM:

The proposed system of women's safety gloves is designed to overcome the shortcomings of existing ones by combining ergonomic fit, protective materials, and smart technology. Unlike traditional gloves that mainly focus on workplace hazards, the new system emphasizes both physical protection and personal safety. These gloves would be tailored to women's hand dimensions for better comfort and dexterity, while using cut-resistant and waterproof materials to handle industrial or household risks. At the same time, they would integrate modern features like an SOS button, GPS tracking, Bluetooth connectivity, and distress sensors to provide real-time alerts in emergencies. Some designs also suggest adding alarm or mild shock mechanisms for self-defense. Overall, the proposed system aims to create gloves that are not only safer and more comfortable for women at work but also act as a discreet personal safety device in everyday life.

3.2.1. Advantages of Proposed System

1 Ergonomic Fit

- Gloves designed using women-specific anthropometric data ensure proper fit.
- Better dexterity and grip strength compared to scaled-down men's gloves.
- Reduced fatigue during long shifts.

2 Enhanced Comfort

- Lightweight and breathable materials improve usability.
- Flexible joints allow natural hand movement.
- Sweat-resistant
- prevent discomfort.

3 Workplace Protection

- Cut-resistant materials protect against sharp tools.
- Waterproof and chemical-resistant layers for industrial and household safety.
- Durable design ensures long-term use.

4 Personal Safety Enhancements

- Built-in alarm system to deter attackers.
- Mild shock mechanism for self-defense.
- Stress or heart rate sensors to detect distress and auto-trigger alerts.

5 Comfort and Health

- Breathable fabrics reduce sweating and irritation.
- Hypoallergenic materials minimize risk of skin allergies.
- Lightweight design allows prolonged use without discomfort.

4 SYSTEM REQUIREMENTS:

4.1HARDWARE REQUIREMENTS:

1.MICRO CONTROLLER ARDUINO ESP32:

- **Library:** The Arduino Matter library is available in the arduino-esp32 package.

- **Compatibility:** Works with **Wi-Fi** (ESP32, ESP32-S3) and **Thread** (ESP32-C6, ESP32H2). can be paired via **QR code** or **manual setup code**.
- **Endpoints:** Pre-built device types include **lights, switches, sensors, plugs**.
- **Ecosystem Integration:** Supports **Apple HomeKit, Google Home, Amazon Alexa** simultaneously.

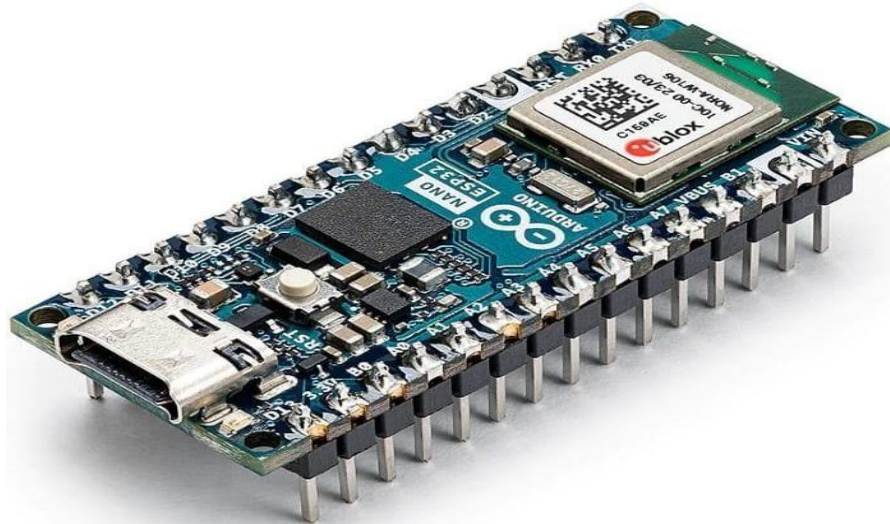


Fig: 1 Micro controller

2.ARDUINO UNO CONNECTED TO GPS MODULE:

An Arduino Uno can be connected to a GPS module (like the NEO-6M) to read location data such as latitude, longitude, altitude, speed, and UTC time. The GPS communicates with the Uno via UART (serial), and the data comes in NMEA sentences that you parse in code.

Arduino Uno R3 (ATmega328P microcontroller) :

- **GPS Module:** Commonly the **NEO-6M GPS receiver**
- **Connections:**
- GPS TX → **Arduino RX (Pin 4 or SoftwareSerial RX)** ◦ GPS RX → **Arduino TX (Pin 3 or SoftwareSerial TX)** ◦ GPS VCC → **5V**
- GPS GND → **GND**
- **Antenna:** The GPS module requires a clear view of the sky for accurate readings

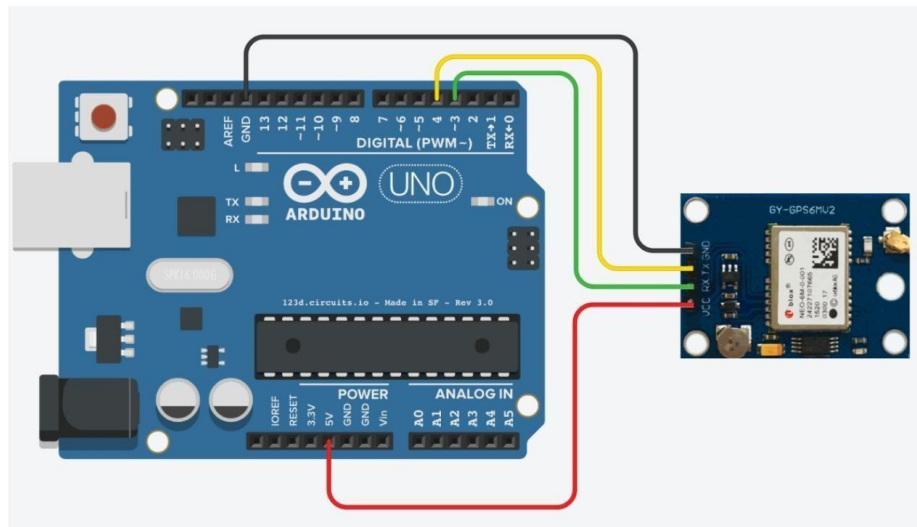


Fig: 2 Arduino UNO connected to GPS module

3.Arduino UNO with GSM SIM800L Module Wiring

This configuration allows the Arduino to communicate with the GSM network via the SIM800L module. You can:

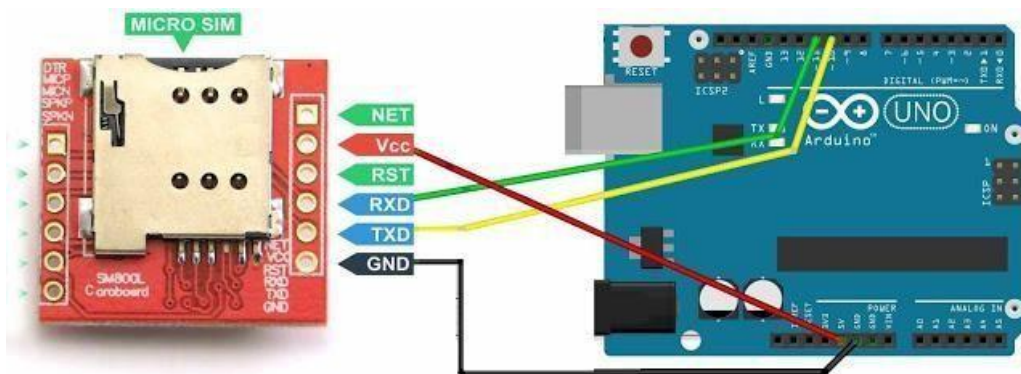


Fig:3 Arduino UNO with GSM SIM800L Module Wiring

- Send and receive SMS messages
- Make and receive phone calls
- Use mobile data (GPRS)

4.2 SOFTWARE REQUIREMENTS:

1.SMS notification system:

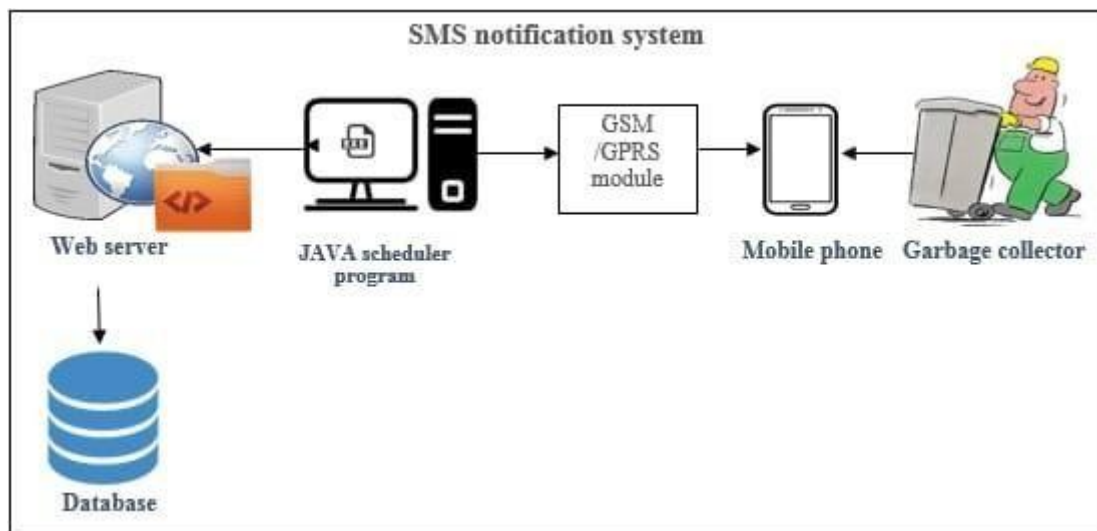


Fig:1 SMS notification system

Reminder: Your appointment is scheduled for tomorrow at 10:00 AM.

- Alert: Your package has been shipped and will arrive by Friday.
- Update: Your account password was successfully changed.

Key Components

- **Message Content:** Short, concise text (usually under 160 characters) that delivers alerts, reminders, or updates.
- **Gateway/Platform:** A service provider or API that connects your system to telecom networks to deliver SMS messages.
- **Database/CRM Integration:** Stores user details (like phone numbers, preferences, and opt-in status) to personalize and automate notifications.

2.ARDUINO IDE

The Arduino IDE (Integrated Development Environment) is the official open-source software used to write, compile, and upload code to Arduino boards. The latest version, Arduino IDE 2.x, offers a modern editor with autocompletion, code navigation, and a live debugger, making it faster and more powerful than the legacy 1.8.x version.

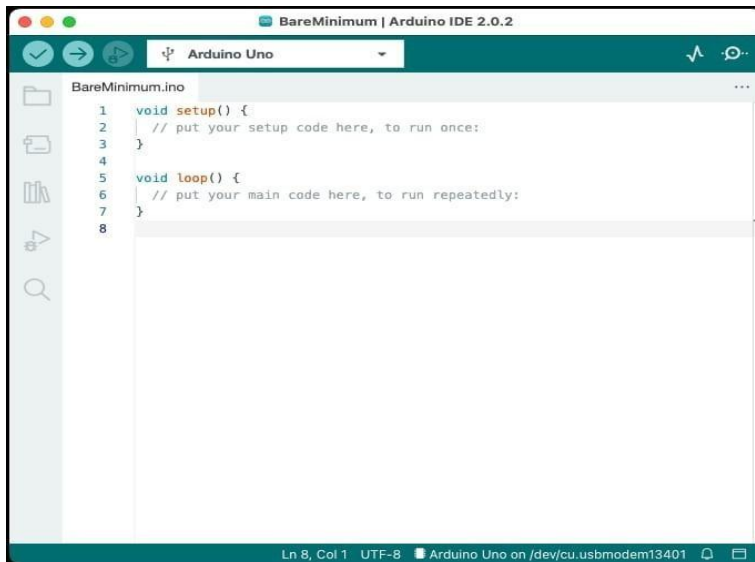


Fig:2 Arduino IDE

What Arduino IDE Does

- **Code Writing:** Supports C and C++ programming languages with built-in libraries.
- **Compilation:** Converts human-written code into machine instructions for Arduino boards.
- **Uploading:** Sends compiled code directly to the connected Arduino board via USB.
- **Debugging:** Newer versions include live debugging tools to help troubleshoot projects.
- **Examples & Libraries:** Comes with ready-made code snippets and libraries for sensors, motors, displays, and more.

5 UML DIAGRAMS

5.1 CLASS DIAGRAM

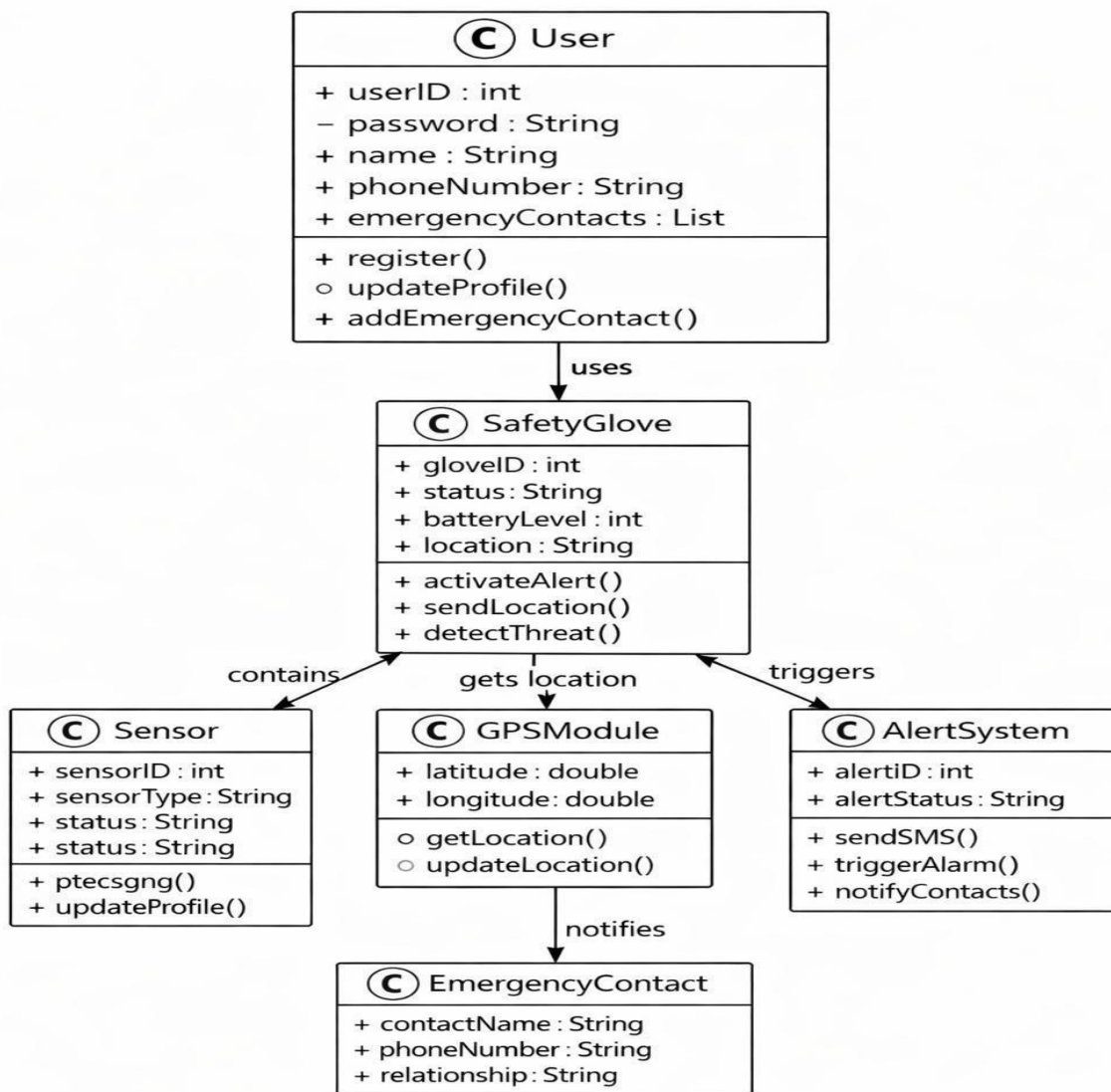


Fig:5.1.1 Class Diagram

A **class diagram** for women's safety gloves explains the system's structure by showing classes, their attributes, and how they relate to each other. The main class is safety glove, which includes attributes such as size, material, fit type, protection level, and sensor integration. The User class represents the female worker, with attributes like hand size, task type, and comfort preference. A Manufacturer class handles design and production, with attributes like design standards and quality control.

5.2.USE CASE DIAGRAM

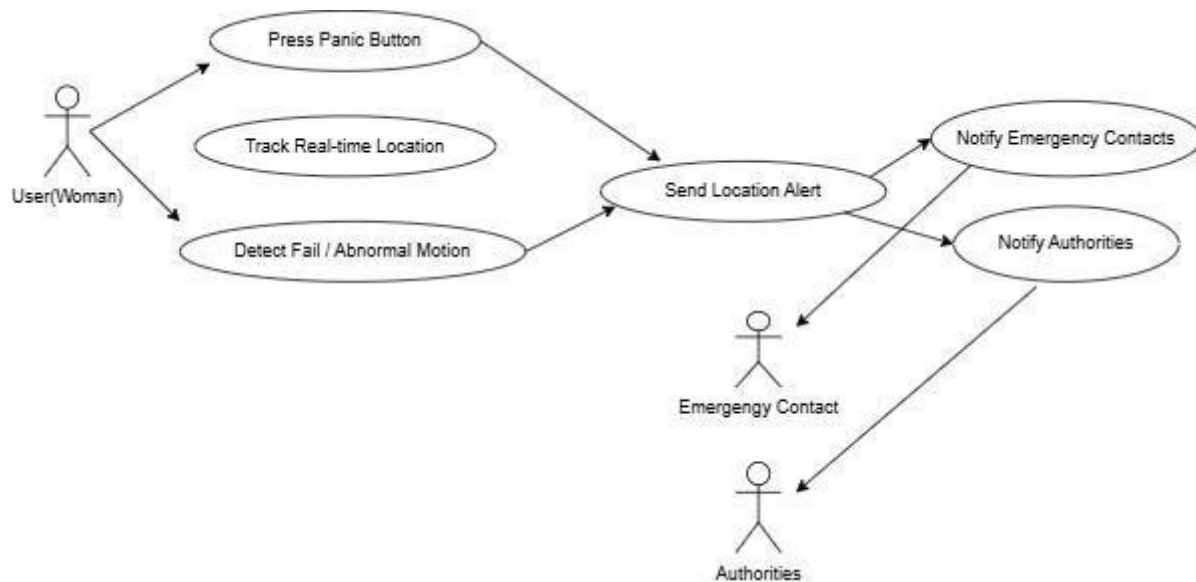


Fig: 5.1.2 Use Case Diagram

A use case diagram for women’s safety gloves would illustrate how different actors interact with the glove system and highlight its functional requirements. In this context, the primary actor is the female worker, who uses the gloves during tasks such as handling machinery, working with chemicals, or performing precision-based operations. Supporting actors could include the safety officer (responsible for monitoring compliance), the manufacturer (who designs and supplies ergonomically fitted gloves), and possibly smart sensors (if the gloves integrate hazard detection technology). The diagram would show use cases like “Wear Gloves for Protection,” “Detect Hazardous Substance,” “Ensure Proper Fit,” “Monitor Safety Compliance,” and “Provide Comfort and Breathability.”

Actors

- **User (Women Worker)** Wears gloves, provides feedback, requests customization.
- **Designer** o Creates glove prototypes using anthropometric data.
- **Tester/Quality Control** o Validates glove safety and performance.
- **Manufacturer**
- Produces gloves based on approved designs.

5.3.SEQUENCE DIAGRAM

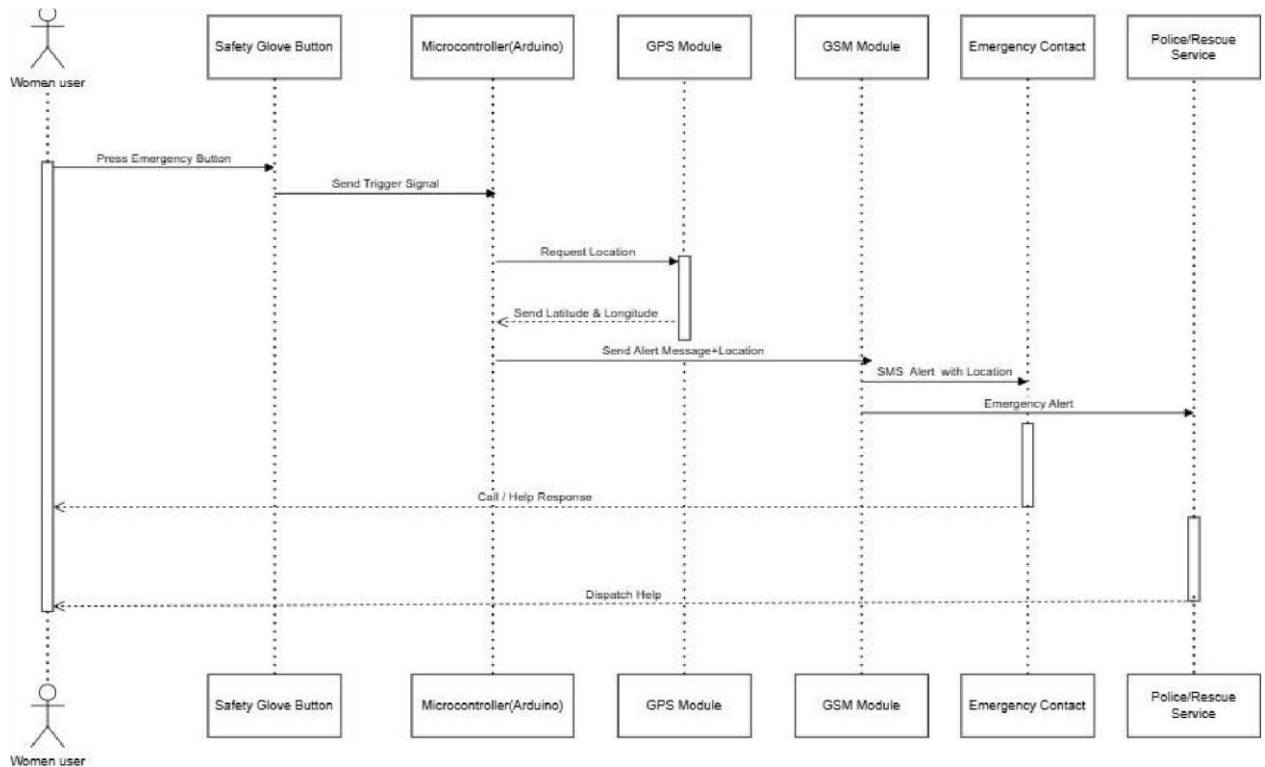


Fig 5.1.3: Sequence Diagram

A sequence diagram is a type of UML (Unified Modeling Language) diagram that visually represents how objects or components in a system interact with each other over time. It focuses on the chronological order of messages exchanged between different entities, such as users, devices, or software modules. In a sequence diagram, vertical lines represent the lifelines of each participant, while horizontal arrows show the flow of communication or actions. This makes it easier to understand the dynamic behavior of a system, especially how processes are triggered and how data moves step by step. For example, in a women's safety gloves project, a sequence diagram could illustrate how the glove detects distress, encrypts the data, sends it to a mobile app, and then forwards the alert to emergency contacts.

Actors & Components

- User (Women Worker)
- Feedback Module
- Design Module
- Testing Module
- Manufacturer

5.4.COMPONENT DIAGRAM

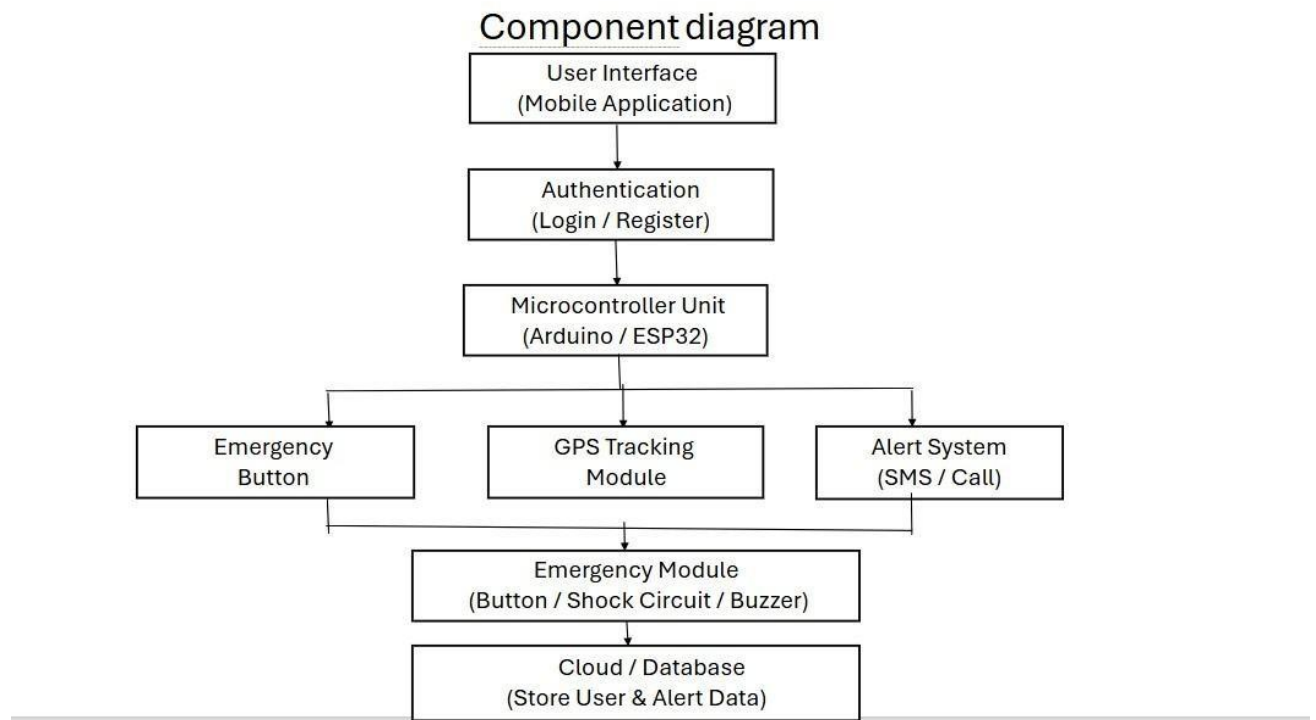


Fig 5.1.4: Component diagram

The component diagram shows how the system is modularized into design, material, testing, feedback, and manufacturing units, all connected through the user interface. This ensures a continuous improvement cycle where women workers remain central to glove design and safety.

5.5 SYSTEM ARCHITECTURE

The system architecture for women safety gloves is designed as a **closed feedback loop** that connects users, designers, testers, materials, and manufacturers. Each component interacts directly to ensure ergonomic fit, safety compliance, and continuous improvement.

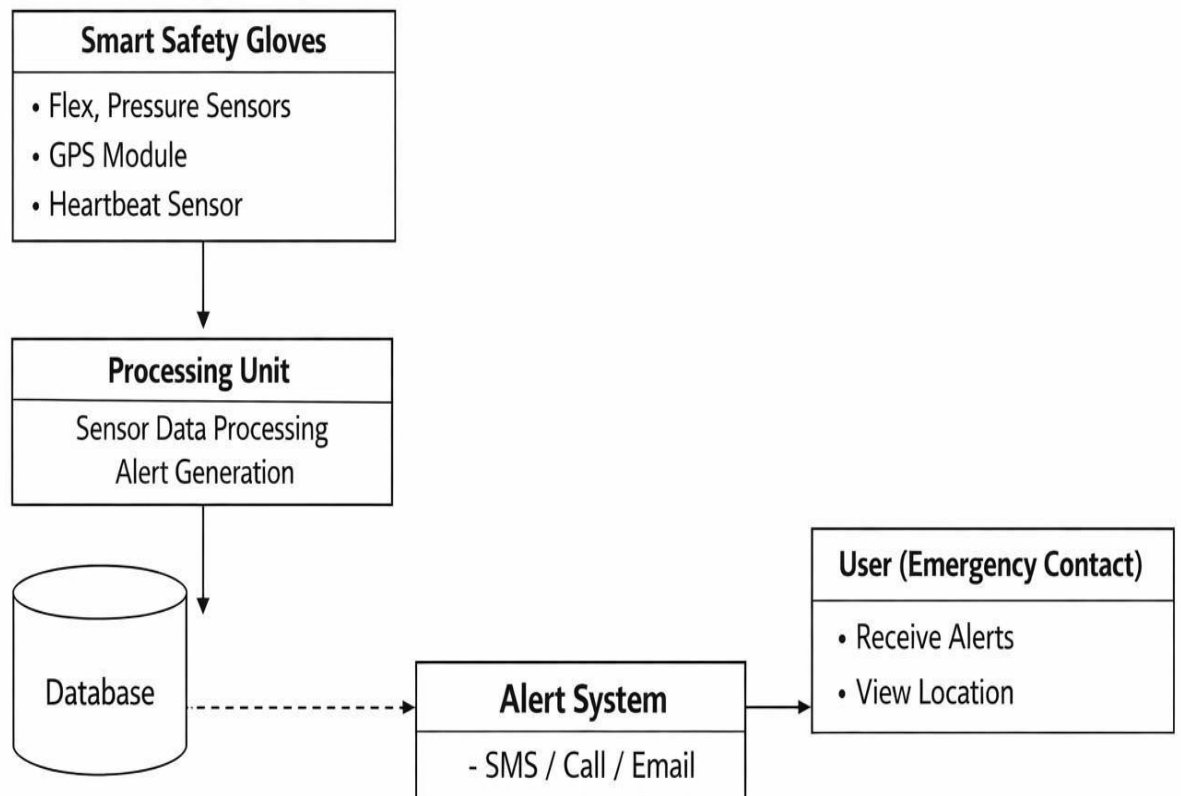


Fig 5.2 :System Architecture

6 .IMPLEMENTATION

6.1 Modules

Encryption:

For a **women's safety gloves project**, encryption ensures that any distress signals or personal data transmitted remain secure and private. The gloves can be designed to detect emergencies and send alerts containing GPS location or biometric information to trusted contacts or authorities. By applying encryption methods such as **AES (Advanced Encryption Standard)** for fast, lightweight data protection or **RSA** for secure key exchange, the system prevents unauthorized access to sensitive information. When paired with secure communication protocols like **TLS/SSL**, the alerts travel through encrypted channels, making interception or tampering nearly impossible.

Decryption:

Decryption is the reverse process of encryption, where encoded data is converted back into its original readable form using the correct key. In your **women's safety gloves project**, decryption ensures that only authorized recipients—such as family members, emergency services, or trusted contacts—can access the distress signals or location information sent by the gloves. For example, if the gloves encrypt an SOS message using AES before transmitting it, the recipient's device or server will use the same key to decrypt the message and reveal the actual alert. This guarantees that sensitive information remains protected during transmission, while still being accessible to those who need it in an emergency.

6.2 SYSTEM TESTINGS

System settings information refers to the configuration options that control how a device, application, or operating environment behaves. These settings typically include parameters for hardware, software, security, and user preferences. For example, in a computer or mobile device, system settings may cover network connections, display resolution, sound levels, storage management, and accessibility features. They also often include security controls such as password management, encryption options, and privacy permissions. By adjusting system settings, users can personalize their experience, optimize performance, and ensure that the device operates securely and efficiently. In projects like smart wearables or IoT devices, system settings define how sensors interact, how data is transmitted, and how alerts are managed, making them essential for both functionality and user safety.

SOURCE CODE:

```
#include <SoftwareSerial.h>
```

```

#include <TinyGPS++.h>

// GSM
SoftwareSerial gsm(10, 11); // RX, TX

// GPS
SoftwareSerial gpsSerial(4, 3); // RX, TX
TinyGPSPlus gps;

// Pins const int
buttonPin = 2; const int
buzzerPin = 8;

// Phone number
String phoneNumber = "+918341737616";

// Button state bool lastButtonState =
HIGH; void setup() {
  Serial.begin(9600); gsm.begin(9600);
  gpsSerial.begin(9600);
  pinMode(buttonPin, INPUT_PULLUP);
  pinMode(buzzerPin, OUTPUT);
  delay(3000);

  Serial.println("System Ready..."); // GSM
  Initialization gsm.println("AT"); delay(1000);
  gsm.println("AT+CMGF=1"); // SMS text mode
  delay(1000); gsm.println("AT+CSQ"); // signal
  check delay(2000); while (gsm.available()) {
    Serial.write(gsm.read());
  } randomSeed(analogRead(A0));
} void loop() { // Read GPS
  continuously while
  (gpsSerial.available()) {
    gps.encode(gpsSerial.read());
  }
  bool currentState = digitalRead(buttonPin);
  // Detect button press if (lastButtonState == HIGH &&
  currentState == LOW) { Serial.println("SOS
  ACTIVATED!") digitalWrite(buzzerPin, HIGH); String

```

```

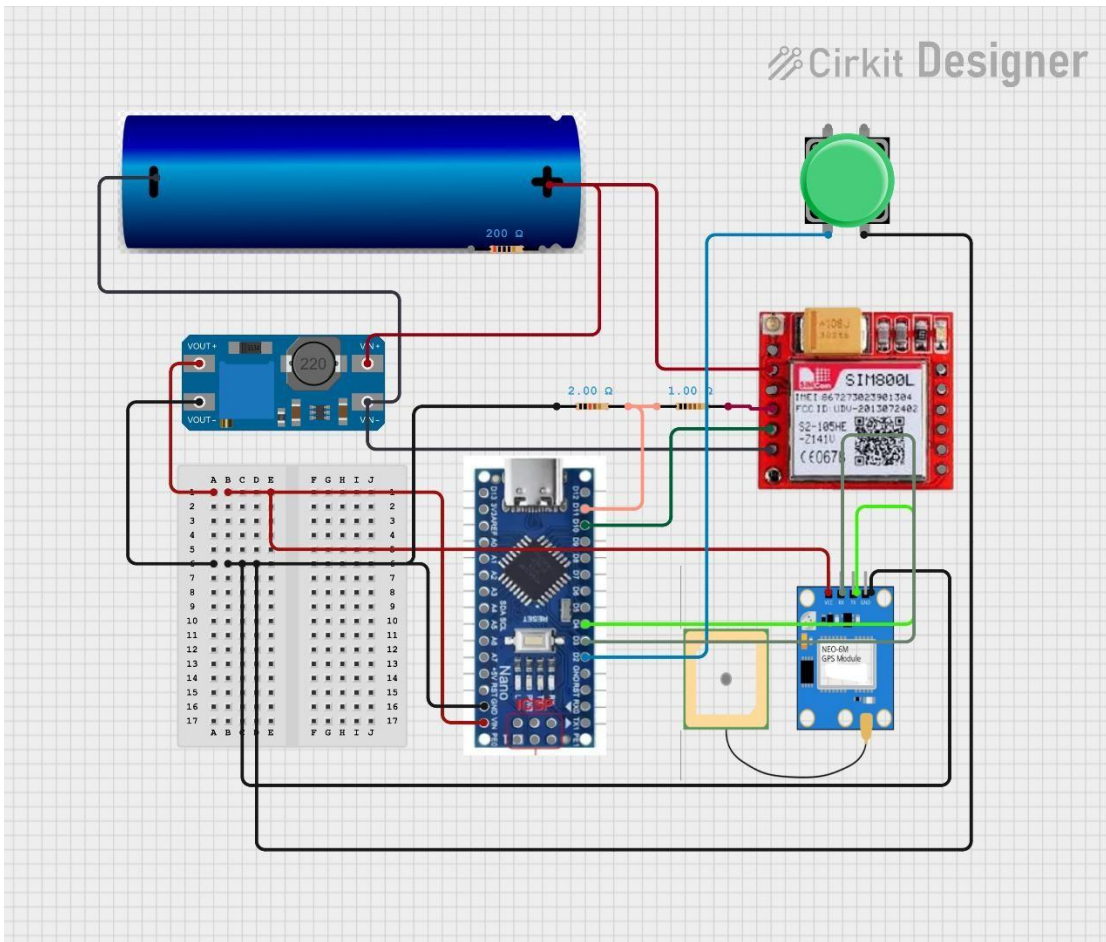
location = getLocation(); sendSMS(location); delay(3000);
// IMPORTANT: wait before call makeCall();
digitalWrite(buzzerPin, LOW);
}
lastButtonState = currentState;
}
// GET LOCATION
String getLocation() { if
(gps.location.isValid()) {
    String link = "https://maps.google.com/?q=";
link += String(gps.location.lat(), 6); link +=
","; link += String(gps.location.lng(), 6);
Serial.println("GPS: " + link); return link; }
else {
    Serial.println("GPS not fixed, using dummy");
// Hyderabad fallback locations int r = random(1, 5); if (r == 1) return
"https://maps.google.com/?q=17.4933,78.3915"; // JNTU if (r == 2) return
"https://maps.google.com/?q=17.4204,78.3272"; // CBIT if (r == 3) return
"https://maps.google.com/?q=17.5185,78.4078"; // OU if (r == 4) return
"https://maps.google.com/?q=17.3850,78.4867"; // Nizam return "Location
unavailable";
}
}
// SEND SMS void
sendSMS(String msg) {
Serial.println("Sending SMS...");
gsm.println("AT+CMGF=1");
delay(1000);
gsm.print("AT+CMGS=\"");
gsm.print(phoneNumber);
gsm.println("\"); delay(2000); //
wait for >
gsm.println("EMERGENCY ALERT!");

```

```
    gsm.print("Location: ");
    gsm.println(msg);
    gsm.write(26); // CTRL+Z
    delay(5000);
    Serial.println("SMS SENT");
}
// MAKE CALL
void makeCall() {
    Serial.println("Calling...");
    gsm.println("AT");
    delay(1000);
    gsm.print("ATD");
    gsm.print(phoneNumber);

    gsm.println(";");
    delay(10000); // call time
    gsm.println("ATH");
    Serial.println("CALL ENDED");
}
```

RESULT



CONCLUSION

The women's safety gloves project can be seen as a groundbreaking initiative that merges wearable technology with security and communication systems to address one of the most pressing social concerns—women's safety. These gloves are not just ordinary wearables; they are designed to act as a discreet yet powerful tool that empowers women in vulnerable situations. By embedding sensors, GPS modules, and communication chips, the gloves can detect distress signals and instantly transmit alerts to trusted contacts or emergency services. The integration of encryption and decryption mechanisms ensures that sensitive information such as location data or SOS messages remains secure during transmission, protecting the wearer's privacy while guaranteeing that only authorized recipients can access the alerts.

Beyond the technical aspects, the project highlights how **IoT and smart devices** can be harnessed for social good. The gloves provide a sense of confidence and reassurance, knowing that help can be summoned quickly and securely. They also demonstrate the importance of combining hardware innovation with robust software protocols, such as lightweight encryption algorithms suitable for lowpower microcontrollers, secure key management, and end-to-end communication channels. Moreover, the project emphasizes scalability— future versions could integrate biometric authentication, health monitoring, or even AI-based threat detection to further enhance safety.

In essence, the women’s safety gloves project is more than a technological prototype; it is a vision of how wearable devices can serve as guardians in everyday life. By blending **functionality, security, and social impact**, the project not only addresses immediate safety concerns but also contributes to building a culture of empowerment and trust. It stands as a reminder that technology, when thoughtfully applied, can be a powerful ally in creating safer environments and fostering independence for women across diverse contexts.

7 FUTURE ENHANCEMENTS

1.GPS INTEGRATION WITH REAL TIME TRACKING:

Embedding a GPS module will allow the gloves to send precise location data during emergencies. This can be linked to a mobile app or cloud platform so family members or authorities can track the wearer in real time.

2.BIOMETRIC SENSORS FOR AUTOMATIC ALERTS:

Adding sensors to monitor heart rate, skin temperature, or stress levels can help detect panic or abnormal conditions. If the glove senses sudden spikes in stress, it can automatically trigger an SOS alert without manual input.

3.VOICE ACTIVATION AND GESTURE CONTROL:

Hands-free activation through voice commands or specific gestures (like clenching the fist twice) ensures alerts can be sent even if the wearer cannot press a button. This adds convenience and accessibility.

4.IOT AND CLOUD CONNECTIVITY:

Connecting the gloves to IoT platforms allows data to be stored and analyzed. Cloud integration can provide advanced features like predictive analytics, history logs, and integration with smart city safety networks.

5.AI BASED THREAT DETECTION:

Machine learning algorithms can analyze sensor data and environmental inputs to predict unsafe situations. For example, detecting unusual movement patterns or identifying high-risk zones based on location history.

6.MULTI CHANNEL EMERGENCY ALERTS:

Instead of relying only on SMS, alerts can be sent via mobile apps, push notifications, emails, and even direct integration with police control rooms. This redundancy ensures messages reach recipients quickly.

7.ADVANCED ENCRYPTION AND SECURE COMMUNICATION:

Strengthening encryption protocols (AES, RSA, TLS) ensures that sensitive data like location and identity remain private. Future versions could use blockchain-based authentication for tamper-proof communication.

8.IMPROVED BATTERY AND CHARGING OPTIONS:

Enhancements could include lightweight rechargeable batteries, solar charging strips, or kinetic charging (using hand movement). This ensures the gloves remain functional for long durations without frequent charging.

9.INTEGRATION WITH OTHER WEARABLES:

Syncing the gloves with smartwatches, fitness trackers, or smartphones creates a connected ecosystem. This allows cross-device alerts, health monitoring, and seamless communication between gadgets.

10.CUSTOMIZED ALERTS AND USER PROFILES:

Users could personalize emergency contacts, alert messages, and sensitivity levels. For example, one profile could send alerts to family, while another could notify workplace security.

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